In the figure, the result consists of *true positives* (relevant documents) as well as *false positives* (irrelevant documents). Precision is about how many of the retrieved documents are relevant. In this case, we know 6 documents are TVs out of 10 resulting documents, so 6 documents are relevant, and the remaining 4 are irrelevant. Hence, the precision is

```
precision = 6 / 10 × 100% = 60%
```

This tells us that only 60% of the documents returned are relevant. Some documents not directly related to the query (irrelevant documents) also appear in the results, for various reasons.

10.1.2 Recall

Recall is the other side of the coin. It measures how many documents that were returned are relevant. For example, there may have been a few more relevant results (TVs) that were not returned (i.e., were omitted) as part of the results set. The percentage of relevant documents retrieved is called *recall* (see figure 10.2).

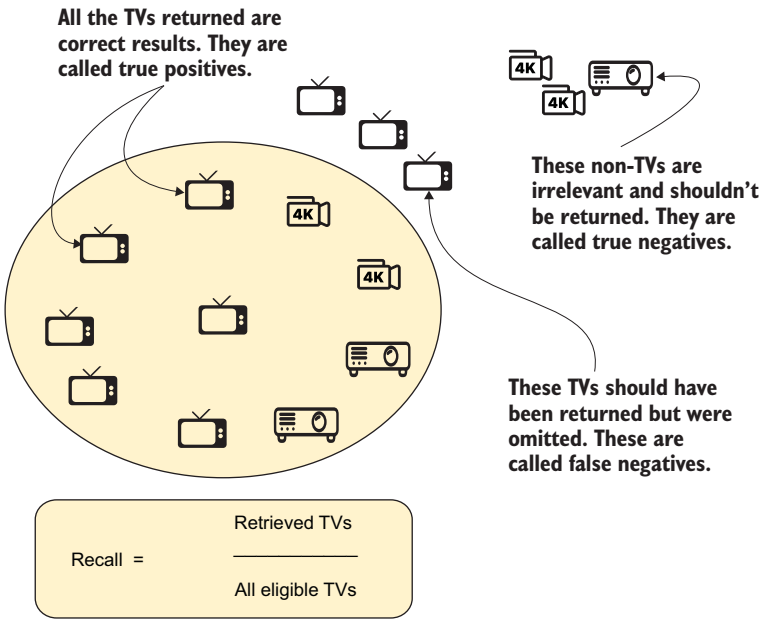


Figure 10.2 The recall measure in action for our 4K television example search

As the figure shows, three TVs fell into the search bucket but were not returned. These are called *false negatives*. On the other hand, a few products like cameras and projectors weren't returned that are genuinely irrelevant and, thus, not expected to